

Group Password Strength Meter Based on Attention Mechanism

Daojing He, Beibei Zhou, Xiao Yang, Sammy Chan, Yao Cheng, and Nadra Guiana

ABSTRACT

User authentication is an important means to ensure the security of users' cyber accounts. Although there are various authentication means such as irises and fingerprints, passwords are still the main authentication method for the foreseeable future due to their low cost and easy implementation. Password strength evaluation is to measure the security strength of passwords, which has been widely studied. However, we found that the current password strength evaluation methods ignore the characteristics from password creators and do not consider the impact of regional groups on password generation. In this paper, we propose the concept of group passwords to analyze the password characteristics of different groups. Based on this notion, a group-based password strength evaluation method is proposed. In addition, we analyze the vulnerabilities of large-scale real-world password groups leaked in previous security incidents. The analysis results show that different password groups have different characteristics. Then, we use the attention mechanism (AM) in the neural network model to learn the dependence between group characteristics and password context features. A long short-term memory (LSTM) model with natural advantages in processing timing features is used to process the password to achieve a more accurate password strength evaluation. We demonstrate the effectiveness of group-based password evaluation using the real-world password data sets.

INTRODUCTION

Password authentication systems are widely used in Internet services because of their low cost, simplicity, and easy implementation. They have been and will continue to be the mainstream authentication method.

In the past few years, large-scale information disclosure incidents have occurred. For example, from 2013 to 2016, account data of about one billion Yahoo users, including names and passwords, were continuously stolen by hackers. In 2014, there was password leakage in China's well known mobile phone manufacturer Xiaomi Corporation, affecting about eight million users. In September 2018, Facebook discovered a security vulnerability that hackers could exploit to obtain information which enables them to

control about 50 million user accounts. In 2018, a well known hotel group in China, Huazhu Hotel, leaked about 250 million pieces of data containing personal information such as user name and ID number due to improper operation of programmers.

Meanwhile, password guessing attacks have undergone rapid development [1]. From the initial heuristic approach, they have been evolved to be completely probabilistic model-based [2], and further evolved to be based on neural network models. For example, Melicher *et al.* [3] proposed a guessing attack using a neural network to simulate passwords. Hitaj *et al.* [4] proposed the use of an antagonistic generation network for password guessing and confirmed its validity. The accuracy and efficiency of their attack algorithm is improved by the use of the large-scale real password data obtained from password leak incidents.

The security of password-based authentication systems relies on the password strength, which further requires users to generate strong passwords. Web service providers also make efforts to enhance their auxiliary system capabilities to assist users creating strong passwords [5]. For example, some web services providers do not allow users to use blacklisted passwords and use Password Strength Meters (PSMs) to measure the strength of password created by users. Web service providers invest heavily in PSMs [6]. They design PSMs based on heuristic algorithms and use the entropy value as the evaluation index. Such PSMs may not be able to accurately characterize the password strength, which makes the user mistakenly think that their passwords are strong but they are in fact vulnerable to modern guessing attacks [7].

To address this problem, we first introduce a concept named group password to accurately characterize the password creation preference of users from different groups. Based on such a concept, we propose a PSM that takes into account password group information. In the remainder of this article, we first discuss the existing mainstream password strength evaluation methods, analyze their advantages and disadvantages, and then analyze the vulnerability of large-scale real password data from the perspective of group password. Then, based on a neural network model and probability model, we propose a group-based password strength evaluation method. Finally, the existing password sets are divided into groups, and comparative experiments are

carried out to verify the effectiveness of the proposed method.

RELATED WORK

By analyzing the leaked real-world password data sets, we are surprised to find that weak passwords still exist in major websites. There are two main reasons for the widespread use of such weak passwords; the user's insufficient security awareness and the inaccurate PSMs used by those websites. The same password could be evaluated to different security levels by the PSMs of different websites. According to the design of the PSMs, we divide them into the following three categories.

Rule-based PSMs. The core idea of the rule-based PSMs is to determine whether the user password meets the password composition rules formulated by the service provider. For example, whether the password is sufficiently long, whether the number of character types is sufficient, etc. The evaluation index of such PSMs is the entropy value (generally Shannon entropy). NIST PSM [8] is one of the well known PSMs of this type. The advantage of rule-based PSMs is that they are simple and easy to implement. However, the disadvantage is that the password strength evaluation result is not accurate.

PSMs Based on Pattern Matching. The core idea of PSMs based on pattern matching is to detect whether there is a preset pattern in the password string. Common patterns are, for example, keyboard layout patterns (such as "qwerty", "awdrgy", etc.), sequential patterns (such as 123456, etc.) and semantic patterns. The strength of a password is evaluated by summing up the scores of all the patterns in the password. A representative of such PSMs is Zxcvbn PSM [9]. In 2019, Han *et al.* [10] proposed an interpretable PSM based on text pattern recognition by analyzing the leaked Chinese password sets. They analyze the Chinese Pinyin and the Chinese double spelling input method for Chinese characters to match the password string pattern. PSM based on pattern matching is more accurate than rule-based password strength evaluation, but it is still a heuristic approach, which is susceptible to subjective factors.

PSMs Based on Attack Algorithms. This type of PSM is to simulate the behavior of the attacker launching a password guessing attack and evaluate the password strength based on the number of guesses. At present, the PSMs based on password guessing attacks are mainly divided into the following three types: PSMs based on password guessing tools, PSMs based on probability models, and PSMs based on neural networks. Password guessing tools are widely used in password guessing attacks. The most common tools are Hashcat [11] and John the Ripper (JtR) [6]. Houshmand *et al.* [12] applied probabilistic context-free grammar (PCFG) to PSMs, and proposed PCFG-Based PSM which evaluates password strength by the probability of the password. Castelluccia *et al.* [13] proposed a Markov-based PSM. With the rapid development of neural networks in recent years, Melicher *et al.* [3] proposed to obtain the probability distribution of the password by using the recurrent neural network (RNN) to guess the password. Then, Amico [14] use the Monte Carlo

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simulation algorithm to obtain the number of password guesses.

GROUP-BASED

LARGE-SCALE REAL PASSWORD DATA ANALYSIS

In recent years, large-scale user passwords leakage incidents have occurred, which seriously threaten the security of users' accounts. On the other hand, from the massive real-world password data sets, we can obtain useful insight about password characteristics. In this section, we analyze multiple real password data sets from group passwords perspective.

DEFINITION OF THE GROUP

Group members interact among themselves, and have the same characteristics such as their needs and goals. The group we define is a group of people with the same characteristics as long as the characteristic can separate the group. There is no mandatory interaction between group members. Users can be divided into different groups. For example, according to gender, they can be divided into male groups and female groups, and according to birthdays, they can be divided into different age groups.

PASSWORD DATA SETS

In order to better characterize users' passwords, and analyze their vulnerabilities, a total of six password data sets of different languages and different service types are used in our study, totaling about 200 million real user passwords.

According to the service types of websites, our study divides the password data sets into two groups.

- The Game group: contains passwords from websites providing game services. The total number of passwords is 19,721,000, and the number of distinct passwords is 10,245,136.
- The Mailbox group: contains passwords from websites providing mailbox services. The total number of passwords is 68,235,411, and the number of distinct passwords is 35,523,147.

Next, we will analyze the group passwords in more detail from the aspects of password structure, popular passwords and personal information.

Group Password Character Structure. Characters are the basic building blocks of passwords. Password character structures also have a large impact on password strength. In addition to being affected by the user's own habits, the password character structure is also affected by the password generation rules of different websites. We explored the effects of different groups on the structure of password characters. We divided the password characters into four types: uppercase letters (U), lowercase letters (L), digits (D), and special characters (S). We analyzed the proportion of nine password compositions in different

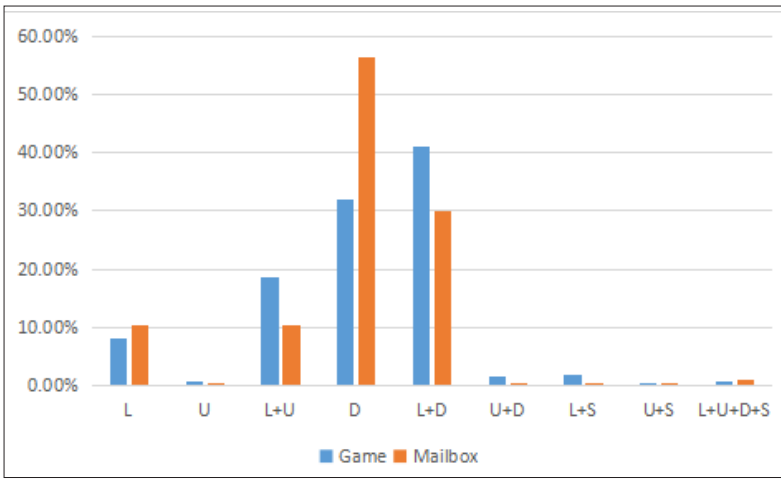


FIGURE 1. Proportion of different character structures.

Ranking	Game		Mailbox	
	Password	Percentage	Password	Percentage
1	123456	1.39%	123456	1.24%
2	111111	0.36%	123456789	0.31%
3	123456789	0.22%	111111	0.23%
4	123123	0.2%	000000	0.17%
5	5201314	0.12%	5201314	0.13%
6	12345678	0.06%	123123	0.13%
7	123321	0.06%	a123456	0.09%
8	7758521	0.05%	7758521	0.07%
9	wangyut2	0.04%	123456a	0.06%
10	1314520	0.04%	12345678	0.05%

TABLE 1. Popular group passwords analysis.

group password, i.e., L, U, L+U, D, L+D, U+D, L+S, U+S, L+U+D+S. It can be seen from Fig. 1 that the structure with the highest proportion of the game group is L+D, while the mailbox group is D. The D structure almost occupies 60 percent of the mailbox group, but the proportion of L+D structure is significantly lower than that of the game group. The proportion of the most complicated password structure L+U+D+S in the mailbox group is 1.10 percent, but it is much higher than that of the game group.

Popular Passwords in Group Passwords. We investigated the top 10 most common passwords in the data sets of the two major groups. As can be seen from Table I, the most popular password among the two groups of users is relatively simple, which is “123456”. It is interesting to note that the popular passwords in the game group are almost all pure numbers, with the exception of “wangyut2”. And the popular passwords in the game group do not have “000000”, which is significantly different from the mailbox group. One of the possible reasons is that “0” represents failure in the game. What is more distinctive in the mailbox group is that “a123456” and “123456a” are popular passwords, and they are not in the

game group. We suspect that this is related to the mailbox password generation policy requiring a mixture of letters and numbers. In addition, we also found some interesting commonalities: “love” plays an important role in the passwords of two groups. For example, the pronunciation of “5201314” and “1314520” are very similar to “I love you forever” and “Forever I love you” in Chinese.

Personal Information in Group Passwords.

The study of personal information in passwords has always been an important part of password analysis. As early as the late 1970s, scholars used personal information to construct password dictionaries to improve the success rate of password guessing attacks. Based on the email account information leaked in the 12306 website, we match mailboxes among different password sets to find passwords that are linked to the same mailboxes which means the same user. We use the regular expression for name string matching detection. Mailboxes, mobile phone numbers, dates, and ID numbers can be detected using regular expressions. Comparing the other two groups, the proportion of personal information in the Mailbox group is significantly lower, which indicates that the users may be more concerned about the security of their passwords. We analyze the usage of personal information in passwords and show the results in Fig. 2. There is an obvious difference in the usage of names and dates in passwords. The passwords in the game group have more dates (26.3 percent). Almost no phone numbers are in the mailbox group.

GROUP PASSWORD STRENGTH EVALUATION BASED ON ATTENTION MECHANISM

At present, almost all websites give certain password generation policies when users register and generate passwords. The password generation policy usually consists of password composition rules and PSM. The websites will feedback the password strength to users according to its own PSM. For example, “123456”, which is known as the “national password”, has entered the blacklist of passwords on many websites and is not allowed to be used. However, we found that there is no uniform standard for measurement, and the research effort on PSM is insufficient, resulting in the situation that the same password receives different feedback results from different websites, which often brings confusion to users.

Because passwords are mainly related to human factors, it is difficult to fully characterize password strength. However, using the high-dimensional function fitting ability of the neural network can better explore the potential connection between password contexts. Based on the research described previously, we can see that different groups show different characteristics and preferences when generating passwords. Therefore, we will use the attention mechanism to obtain the password context dependency of different group characteristics. Then, we combine the RNN to build the neural network model, and use the model to calculate the probability distribution of the passwords of different groups, and finally evaluate the group password strength.

An RNN is a class of artificial neural networks where connections between nodes form a directed graph along a temporal sequence. Long short-term memory (LSTM) is an artificial RNN architecture used in the field of deep learning, which was proposed by Hochreiter and Schmidhuber [3]. Unlike standard feed forward neural networks, LSTM has feedback connections. It can not only process single data points (such as images), but also entire sequences of data (such as speech or video). The concept of time-series in RNN provides natural advantages in dealing with sequence problems. The original RNN has the problem of vanishing gradients. The LSTM model with gate concept solves this problem well. In recent years, the attention mechanism has been widely used in the fields of image, speech and natural language processing. Ma *et al.* [15] proposed to add attention mechanism to the combination of LSTM and target vector, so that the network model pays attention to the content related to the entity in the text. This work proves the effectiveness of attention mechanism in text meaning mining.

This section presents the group PSM (AM LSTM) built upon the LSTM model combined with the attention mechanism, which uses the advantages of LSTM for text sequence processing and the attention mechanism to model password sequences. The model simulates the password generation of different groups, and evaluates the strength of the password by the probability output by the model. The greater the probability value is, the weaker the password strength is. The password strength evaluation process is shown in Fig. 3. During the offline model training, AM LSTM PSM encodes group information as well as the

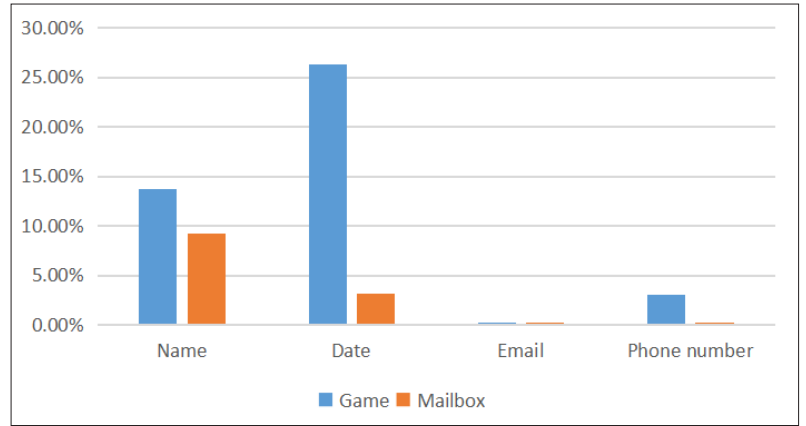


FIGURE 2. Personal information in group passwords.

passwords into vectors. After the completion of the model training, AM LSTM PSM is ready to evaluate given passwords.

AM LSTM MODEL STRUCTURE

AM LSTM mainly consists of four parts: 1) the input layer, which encodes the pre-processed text passwords; 2) the password feature learning layer, which learns user password context feature information; 3) the group weight attention layer, which encodes and vectorizes the group information and at the same time interacts with the password sequential features learned from the previous layer so that the password strength evaluation model pays attention to the group to which the password belongs; and 4) the output layer, which obtains the password probability distribution by nonlinear activation function after the fully connected network.

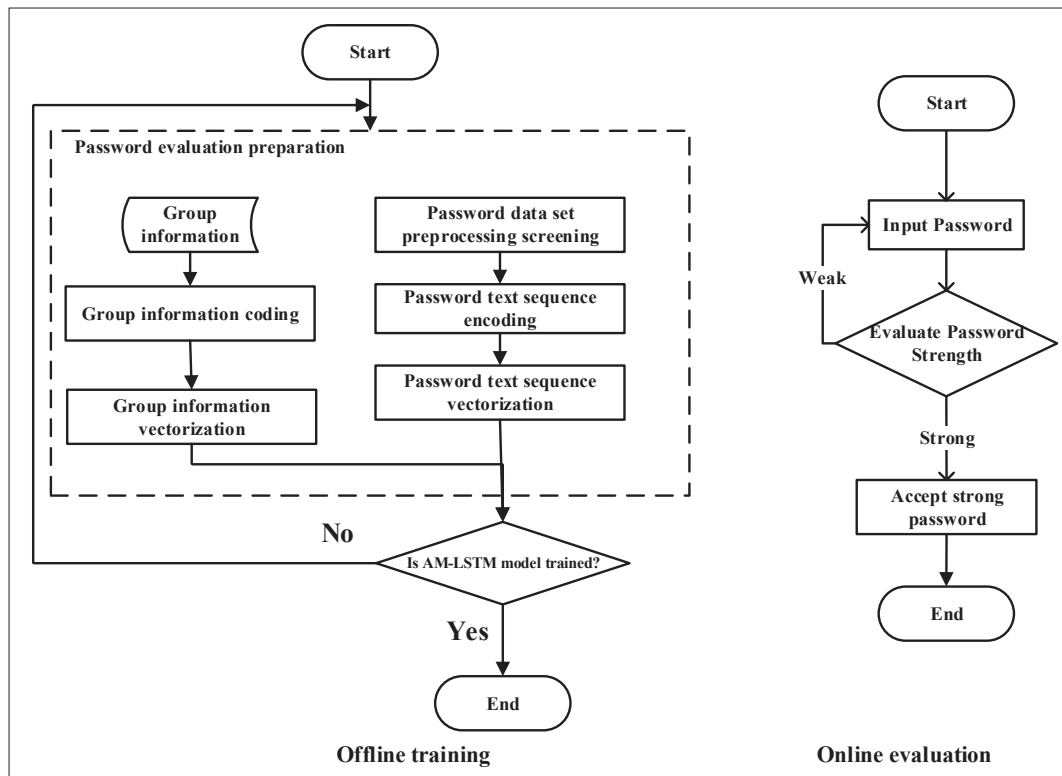


FIGURE 3. Password strength meter process.

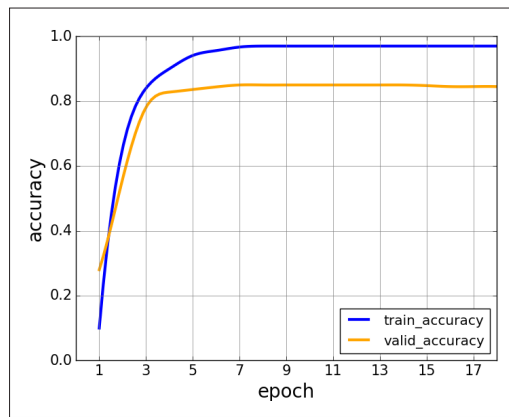


FIGURE 4. Model training process.

Model Input Layer. The password set is pre-processed to obtain valid passwords that contain printable ASCII characters and have valid lengths. Then a text password $w=\{w_1, w_2 \dots w_n\}$, where n represents the length of the password, is converted to one-hot embedding as the input to our AM LSTM model. One-hot code has as many bits as there are states, and only one bit is 1, and the others are all 0.

Password Feature Learning Layer. The object of our study is the text password, which has time series characteristics. That is, the text password is a sequence formed in chronological order, so we choose the RNN in the network model selection. Because the original RNN has the problem of vanishing gradients, we choose the RNN model with the gate concept — LSTM to learn the context-dependent feature information in the password. The gate selects the memory or forgetting of the information mainly through a *sigmoid* non-linear function and a vector dot product. One of the LSTM contains three doors (forgotten gate, input gate, output gate) and a memory unit.

Group Weight Attention Layer. Although LSTM can learn the inter-character-dependent feature information in the password text sequence, it does not recognize the degree of influence of different group features on the weight of the text password sequence. We choose the attention mechanism in deep learning to help the password strength evaluation model to focus on the degree of contextual association between different group features and text password sequences. Like the text password sequence, we first precode the group information. The group is numbered, for example, the Game group (0), the Mailbox group (1). Then, embedding is performed by combining the state vector dimension information of the password feature learning layer to generate a text password character context weight sequence combined with the state vector. Like the input layer, the group information variable is formally expressed: $g_i, g_j \in G$, where G is the population set; then the generated password character sequence weight is the dot product with the hidden layer state vector, and finally output to the next layer.

Model Output Layer. In the output layer, the attention-output is passed to the fully connected network to generate the probability distribution of the password space character using the softmax activation function. Then, the maximum probabili-

ty value of the password character is taken as the output.

EXPERIMENT

We conduct experiments to compare AM LSTM PSM with several well known PSMs, i.e., PCFG PSM [12] and NIST PSM [8]. PCFG PSM is an adaptive PSM. When the user generates a weak password, it suggests an alternative password for the user. NIST includes heuristics based on length and compliance to a composition policy on entropy estimation.

In order to evaluate the effectiveness of the PSMs, a commonly acknowledged method is to attack the passwords that are evaluated as strong by those PSMs. We selected the Markov based attack proposed by Claude *et al.* [13] to guess the passwords of the three PSMs. The number of guesses is used as an indicator to determine the effectiveness of each PSM. The larger the number, the stronger the PSM.

Experimental Data Set Settings. The data set used in this experiment is the two major group password sets mentioned previously. Because the experimental process is mainly divided into two parts, model training and evaluation, the data are randomly divided into two parts. In the network model, we divide the data in the ratio of the training set: test set = 4:1. We choose Python3.6 as the programming language and use the neural network framework Keras for model building.

Model training environment:

Operating system (OS): Windows 10 64-bit

Running memory (RAM): 16GB

GPU: GTX 1080Ti

Model main parameters:

LSTM layer number: 3 layers Single

LSTM neuron number: 256

LSTM *time steps*: 10

Loss function: categorical crossentropy (cross entropy)

Optimizer: sgd (random gradient descent)

Learning rate: 0.0001

Batch size: 32

Since the unknown model will converge after training for many rounds, we adopt the strategy of Early Stopping. The core idea is to utilize the accuracy of the validation dataset of each round. The accuracy rate of the current round is compared with the maximum accuracy value in the past 10 rounds. If the accuracy of 10 consecutive rounds is not improving, the training is stopped. If the current accuracy is higher than the maximum accuracy, the maximum accuracy value is updated and set. The maximum number of training rounds is 100, and the training results are shown in Fig. 4.

In Fig. 4, the abscissa represents the number of rounds of training, and the ordinate represents the accuracy. It can be seen that the convergence speed of the model is faster. After seven rounds, the accuracy of the test set does not increase, which indicates that the model has converged.

Experimental Result. From Fig. 5, we can see that the password guessing success rate of the AM LSTM PSM is significantly lower than that of the NIST PSM under the same number of guesses. For the NIST PSM, the number of successes

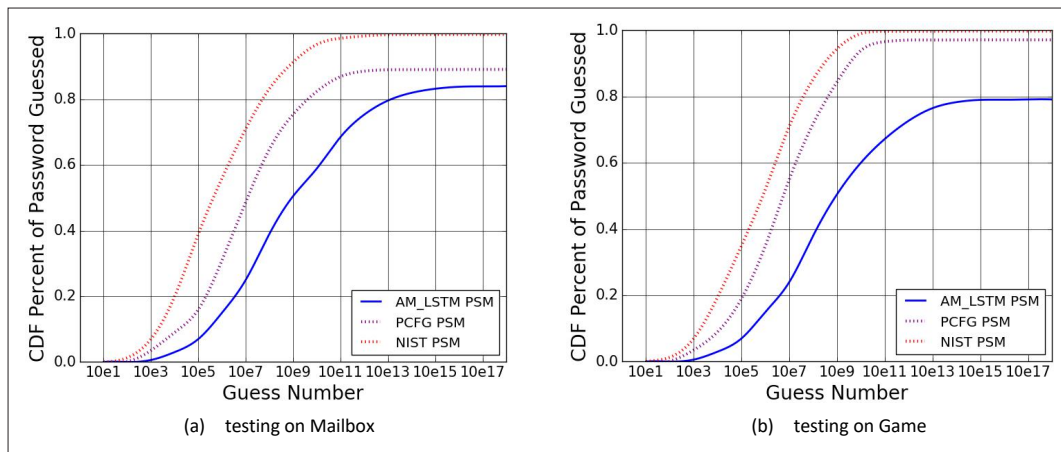


FIGURE 5. Password against guessing attack result graph.

becomes steady when the number of guesses reaches 1011. However, for AM LSTM PSM, the number of guesses needs at least 1015, indicating that the passwords evaluated by AM LSTM PSM are more resistant to guessing attacks. This shows that AM LSTM PSM can more accurately describe the strength of a password than other PSMs.

CONCLUSION AND FUTURE WORKS

Based on the group password concept, we have analyzed a large-scale real password set from the group perspective, including password structure, popular passwords, and personal information. We have demonstrated that there are obviously different characteristics among different groups of passwords. Based on the analysis, we have proposed AM LSTM PSM that takes group characteristics into consideration. AM LSTM PSM uses the attention mechanism to take into account the influence of group characteristics on password context modeled by a long and short term memory network. AM LSTM PSM is able to better evaluate the password strength of different user groups and enhance the password resistance against the guessing attack.

Although this article considers the factors affecting password strength evaluation as much as possible, we believe that it can be further studied from the following aspects:

Group Diversity. We only divide the group according to the purpose of the passwords. However, the group can be divided from multiple dimensions, such as division by age and division by gender. Finding the characteristics of different groups will help to further adjust the parameters of the model, so that the strength evaluation performance is enhanced.

Increase the Amount. The amount of data directly affects the analysis and evaluation results, so we need to collect more password sets for better results.

Analysis Dimension Diversification. We now only analyze password features in terms of character characteristics, popular passwords, and personal information contained. In the future, we can analyze group passwords in more ways, such as length, most frequently appearing numbers, most frequently occurring letters, and so on.

The Model Used. We use the attention mechanism and LSTM model, but the model used is relatively primitive. As the field of neural networks

is developing rapidly, more effective models are emerging and they could be considered to be applied in our work.

More PSM Comparison. We are now only using NIST PSM and PCFG PSM to compare our methods. If we use more PSMs for comparison, our results will be more convincing.

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